



PharmaExpert - MedicationLOT

Concept

Roadmap:

1. **Develop a working interface logic** – ensure the DApp has a functional and user-friendly interface.
2. **Create a working wallet extension app** – for easier interaction with the DApp.
3. **Traceability and authentication** – enable survey and verification of medication parcels, from validation to sale.
4. **Train personnel** – educate staff on the use cases and operation of the DApp.
5. **Build and own infrastructure** – maintain control over the technical and operational backend.
6. **Integrate electronic chipsets** – enhance tracking, authentication, and security.
7. **Enable public/private access views** – allow enterprises and pharmacies to integrate and use the application in real-world scenarios.
8. **Expand the ecosystem** – beyond pharmaceuticals, the system can:
 - Control access to facilities based on eligibility (e.g., for doctors).
 - Monitor and scan user health status and their interactions within labs or facilities.

Impact of Grant Funding on Implementation and Integration

1. Small-Scale Implementation (Pilot Phase)

- **Hardware acquisition:** Electronic chipsets or tracking devices for medication parcels.
- **Development costs:** Paying for the wallet extension, DApp interface, and backend setup.
- **Training personnel:** Educating staff on using the application effectively.
- **Monitoring and testing:** Collecting feedback and ensuring traceability works in real-world conditions.

2. Large-Scale Implementation (Scaling Up)

Once the pilot proves successful, grant funding can accelerate broader deployment:

- **Infrastructure expansion:** Hosting servers, integrating multiple locations, and maintaining high uptime.
- **Regulatory compliance:** Ensuring the system meets health and safety regulations in different regions.
- **User onboarding:** Supporting more pharmacies, hospitals, and labs to join the ecosystem.
- **Ecosystem growth:** Expanding beyond pharmaceuticals, for example to facility access control or health monitoring.
- **Integration support:** Funding partnerships with technology providers, chip manufacturers, and local enterprises.

3. Strategic Advantages

- **Risk reduction:** Grants reduce the financial risk of testing innovative features.
- **Faster adoption:** With funding, you can incentivize early adopters and ensure smoother integration into real-world workflows.
- **Data and analytics:** Grants allow you to build the analytics needed to track medication, user interactions, and system efficiency at scale.

Resources

1. Technical Resources

a. Software & Development Tools

- **Blockchain framework:** Hardhat, Truffle, or Remix for smart contract development.
- **Programming languages:** Solidity (smart contracts), JavaScript/TypeScript (frontend/backend), Python (optional for analytics).
- **Wallet & dApp integration:** MetaMask, WalletConnect, or custom wallet extensions.
- **Databases:** Off-chain storage for messages, logs, and analytics (e.g., Firebase, MongoDB, PostgreSQL).
- **Cloud services:** Hosting servers for the DApp backend (AWS, Azure, or Google Cloud).
- **Testing & simulation tools:** Ganache, Hardhat network, or testnets like Sapphire Testnet or Oasis Testnet.

b. Hardware

- **Electronic chipsets & RFID/NFC tags** for tracking medication parcels.
- **Computers & mobile devices** for testing the wallet and DApp interface.
- **Optional IoT devices** if extending to access control or health monitoring.

2. Human Resources

- **Developers:** Blockchain dev, frontend and backend developers.
- **UX/UI designers:** To make the DApp interface intuitive and user-friendly.
- **Trained personnel:** Staff for testing, using, and demonstrating the system (pharmacy staff, lab technicians).
- **Project manager:** To coordinate development, deployment, and feedback loops.

3. Financial Resources

- **Grant funding** or seed investment to cover hardware, software licenses, and salaries.
- **Operational budget** for pilot testing, marketing, and onboarding users.
- **Maintenance budget** for server costs, blockchain transaction fees (gas), and upgrades.

4. Knowledge & Documentation

- **Blockchain and Solidity tutorials** for team skill-building.
- **Regulatory guidelines** (pharmaceutical tracking, data privacy).
- **Project documentation:** system architecture, workflows, testing protocols.

5. Community & Partnerships

- **Pharmacies, clinics, and labs** for pilot implementation.
- **Technology partners** for chipsets, sensors, or wallet integration.
- **Advisors** for compliance, blockchain scalability, and user adoption strategy.